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Key Points:

- A physics-based model quantifying individuals' vulnerability to combined floodwater and wind hazard is developed
- Winds substantially influence human vulnerability to coastal flooding
- Human physical vulnerability in New York City to coastal flooding during Hurricane Sandy is revealed

Supporting Information:

- Supporting Information S1

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Physical Instability of Individuals Exposed to Storm-Induced Coastal Flooding: Vulnerability of New Yorkers During Hurricane Sandy

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Abstract Flood risk assessment requires a quantitative understanding of hazards and vulnerability. In the coastal built environment, the human's vulnerability to combined hazards due to the floodwater and winds is an integral component of flood risks. The present study aims to reveal the human vulnerability to storm-induced coastal flooding, focusing on New York City during Hurricane Sandy. We develop a physics-based model to quantify individuals' physical vulnerability, both adults and children, to compound hazards of floodwater and winds. The model accounts for the failure of individuals caused by physical instability due to slipping and toppling. The governing equations consider the balance between the driving and resisting forces and moments applied to an individual concurrently exposed to floodwater and winds. We first calibrate the model using existing measurements in the literature and then implement it to study the vulnerability of New York residents in Manhattan to coastal flooding during Hurricane Sandy. Model results indicate that when combined floodwater and wind hazards were at their highest-level during Sandy, the majority of flooded areas were mainly a hazardous zone for adults and either a failure or drowning zone for children. About 5.4% and 47.4% of the total flooded area became a failure zone, and 19.9% and 42.4% became a drowning zone for adults and children, respectively. We conclude that winds can have a significant impact on the physical instability of individuals. For example, model results for children show that neglecting winds results in a reduction of 97.7% in the area of the failure zone.

1. Introduction

Low-lying coastal areas in New York City (NYC) are prone to coastal flooding induced by tropical and extratropical cyclones. Examples of some of the worst historical flood events are Hurricane Donna in 1960, Hurricane Sandy in 2012, and the Great Appalachian Storm of November 1950. Hurricane Sandy generated the highest water level at New York Harbor since, at least, the past 300 years (P. Orton et al., 2019). In NYC, Sandy caused 43 fatalities and \$19 billion in damages (City of New York, 2013). Previous studies have comprehensively revealed hazards associated with Hurricane Sandy (e.g., Lin et al., 2016; Marsooli et al., 2017; P. M. Orton et al., 2016). The present study aims to reveal the vulnerability of individuals in NYC to floodwaters and winds during Sandy.

In addition to hazard mitigation, vulnerability reduction is a core element of flood risk management in the built environment. Flood risk assessment is of great value, specifically in urbanized coastal settings such as NYC, where coastal communities, assets, and infrastructures are highly prone to flooding. Risk assessment requires a thorough understanding of vulnerability, which refers to the “propensity of exposed elements to suffer adverse effects when impacted by hazard events” (Allen et al., 2012; Janssen et al., 2006; Saleh et al., 2017; UNDRO, 1980). In the context of flood risks in urban settings, human vulnerability to floodwaters, due to physical instability and drowning, is a crucial vulnerability assessment component. Advancing understanding of human vulnerability to floodwaters requires predictive tools to identify hydrodynamic conditions that result in the physical instability of individuals with diverse physical characteristics.

Observations show that slipping and toppling are the primary mechanism of physical instability in floodwaters (Jonkman & Penning-Rowsell, 2008; Martínez-Gomariz et al., 2016; Milanese et al., 2014; Xia et al., 2014). Jonkman and Penning-Rowsell (2008) reported that the instability of an individual exposed to floodwaters could be related to two physical mechanisms: friction instability (slipping) and moment instability (toppling). Slipping or friction instability may occur when the drag force applied to the body exceeds the frictional force between the feet and the ground. Toppling or moment instability may occur

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when the moment of forces exerted on the body exceeds the moment due to the body's resultant weight. While high-velocity shallow waters typically cause slipping, toppling is more frequent in deeper waters (Cox et al., 2004; Xia et al., 2014). When the water depth exceeds a person's threshold height (e.g., chin), floating or drowning could occur.

Experimental and numerical studies have investigated the critical (threshold) hydrodynamic conditions that result in different modes of instability in floodwaters. A series of past experimental studies investigated critical combinations of floodwater depth and velocity that result in instability of individuals exposed to floodwaters (Abt et al., 1990; Karvonen et al., 2000; Lind et al., 2004; Russo et al., 2013; Xia et al., 2014). For example, Karvonen et al. (2000) proposed an empirical equation to estimate the critical depth-velocity product based on an individual's height and mass. They derived their equation by conducting laboratory experiments on human subjects in a specific range of age, height, and weight. Xia et al. (2014) conducted a series of experiments in a laboratory flume to obtain the critical water depth and velocity that initiate the slipping and toppling instability of a human body model. They proposed empirical equations to estimate the critical water velocity that results in slipping or toppling instability. The empirical equation estimates the critical velocity for a given flood depth, water density, and individual's height and mass. The equation contains a series of empirical coefficients that were calibrated based on the laboratory experiments.

Empirical equations of human instability are widely used to predict the human vulnerability to flood hazards. For example, the HEC-LifeSim modeling system (CEIWR-HEC, 2020) determines the stability of pedestrians exposed to flooding based on empirical hazard regime curves from Shand et al. (2011). HEC-LifeSim estimates potential life loss and economic damages from flood events, based on comprehensive procedural steps that account for human physical stability, structure and vehicle stability, shelter availability, property damage, flood warnings, issued evacuation time, road networks, among others. The stability of individuals exposed to a given floodwater speed and depth is determined based on stability curves fitted by Shand et al. (2011) to laboratory measurements.

There are some limitations to the existing empirical equations for application to a real-world flood event. While the different empirical equations similarly show that the critical floodwater velocity (regardless of the instability mode) decreases with increasing water depth, they show a significant scatter in the magnitude of critical flood water depth and velocity (Chanson & Brown, 2018; Milanese et al., 2016). The significant scatter in empirical data is due to the different human dimensions and environmental conditions considered in each experimental study. For example, some experiments were conducted with a trained stuntman (Jonkman & Penning-Rowsell, 2008), others with young adults (Martínez-Gomariz et al., 2016), while some were with a model of the human body (Xia et al., 2014). Another limitation with empirical equations is that these equations include a series of empirical coefficients calibrated based on data generated in the laboratory under an idealized and limited range of human dimensions and environmental conditions. Application of these equations to real-world studies where, for example, hydrodynamic conditions widely change in space and time could lead to erroneous interpretation of human vulnerability to flood hazards.

To complement experimental studies, several analytical studies have introduced physics-based conceptual models to relate the individual's instability to flood water depth and velocity (Lind et al., 2004; Milanese et al., 2014; Xia et al., 2014). For example, Xia et al. (2014) and Milanese et al. (2014) developed conceptual models of human instability for slipping and toppling failure mechanisms and drowning. Slipping instability was modeled based on the equilibrium of forces acting at the human body. Toppling instability was modeled based on the equilibrium of the moment of forces applied to the body. The physics-based approach has also been adopted in commercial software such as Life Safety Model that estimates the risk to people posed by dam failure and flooding (Lumbroso et al. 2011). The instability of people is determined based on the floodwater velocities and depths.

The past empirical and physics-based conceptual models of human instability were developed with a focus on inland flooding such as flash and river floods. These models could under- or over-estimate the physical instability of individuals exposed to coastal flooding (induced by storm surge). In contrast to inland flooding, human stability during a coastal flood event can be affected by strong winds. The wind speed during a major hurricane can reach 250 km/h or higher. Depending on the wind direction, relative to the water flow direction, the wind drag force would support or act against the stability of an individual exposed to floodwaters. Analysis of human instability in coastal flood waters should account for the wind effects.

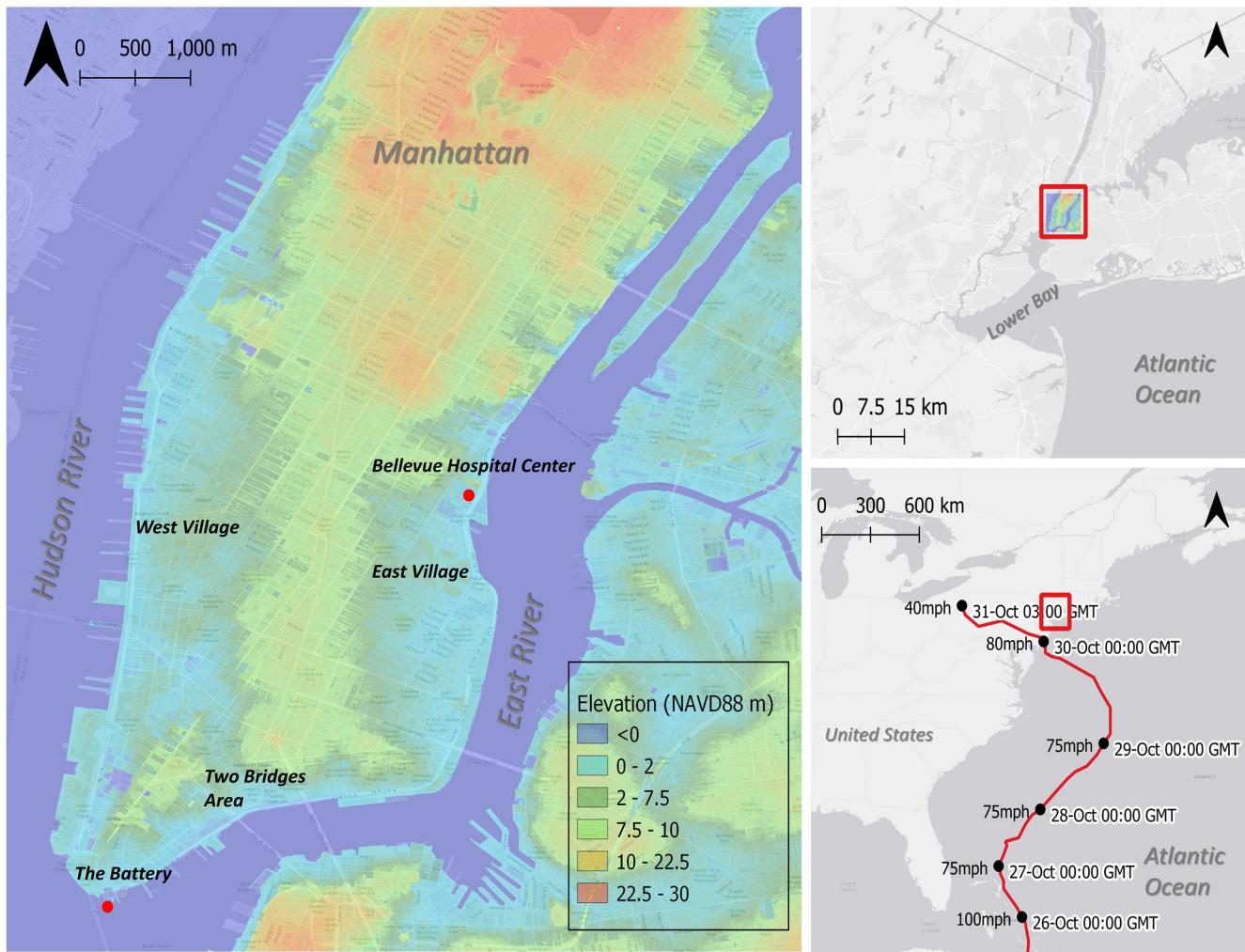


Figure 1. Study area in Manhattan and the storm track of Hurricane Sandy.

The present study aims to reveal the human vulnerability to coastal flooding induced by windstorms. To achieve the study goal, we develop a physics-based model that quantifies the physical instability of individuals who are concurrently exposed to floodwaters and winds. The human instability model, along with hydrodynamic and wind data simulated by a building-scale coastal ocean model, are utilized to determine the spatial and temporal variation of the human vulnerability level in the borough of Manhattan in NYC during Hurricane Sandy.

2. Methodology

2.1. Study Area and Hurricane Sandy

The study area is in Manhattan, including the Midtown and Lower Manhattan, which is the most densely populated borough of NYC. Midtown Manhattan is the world's largest central business district and accommodates some of the world's most marketable real estate. Lower Manhattan is the heart of the United States financial market and home to Wall Street and the World Trade Center. Large areas in Manhattan have a land elevation of below 10 m above the mean sea level (Figure 1), exposing the population and assets in these areas to potential flood hazards.

Manhattan is prone to coastal flooding induced by both tropical and extratropical cyclone storm surge. An example of a damaging historical extratropical cyclone event is the Great Appalachian storm of November

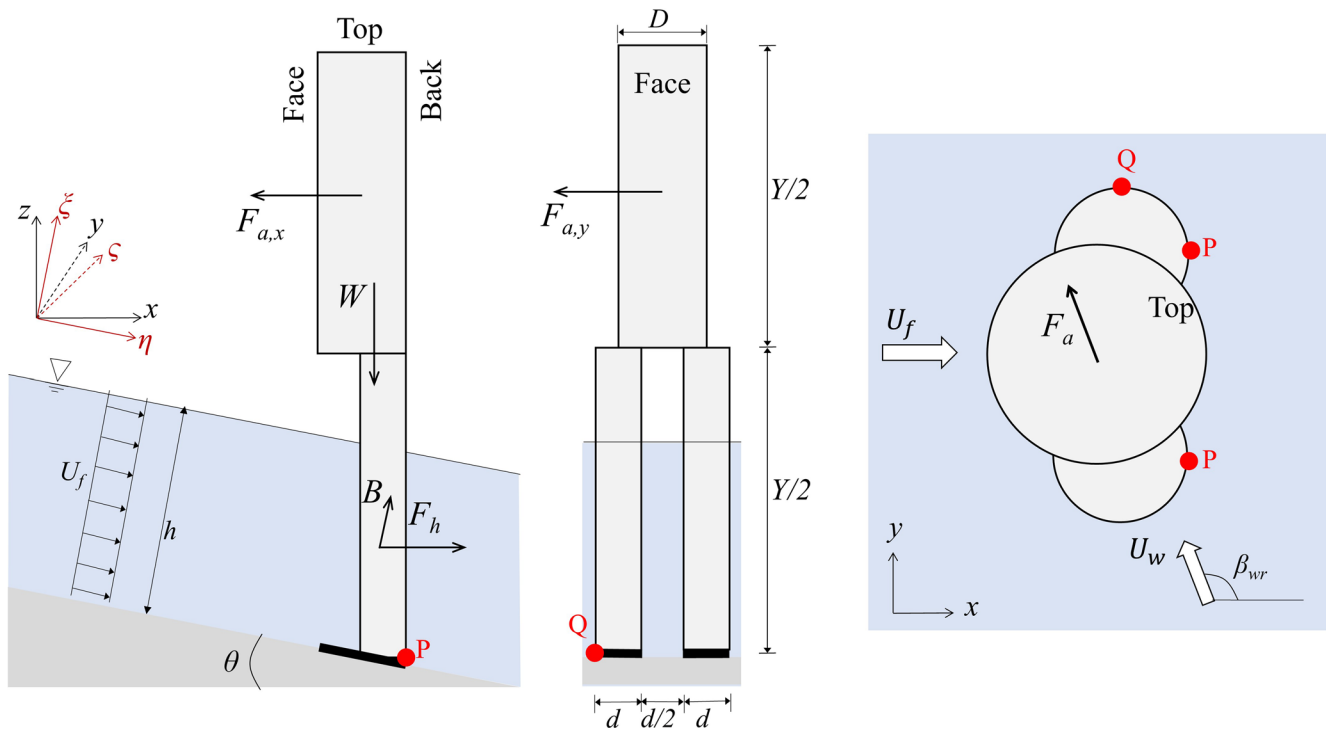


Figure 2. Simplified representation of the human body (left: side view, center: frontal view, right: top view).

1950, which generated a peak surge height of about 2.4 m at The Battery in the southern tip of Lower Manhattan (Colle et al., 2008). An infamous example of a tropical cyclone storm surge event is Hurricane Sandy in 2012, which caused a storm surge of 2.8 m at The Battery.

Hurricane Sandy was the most destructive and costliest hurricane in the northeast United States. At its peak intensity, Sandy was a Category 3 hurricane. It went through an extratropical transition a few hours before making landfall at 2330 UTC on October 29 near Brigantine, New Jersey. Sandy caused 65 billion dollars in damage and 157 deaths in the United States (Blake et al., 2013), with 43 of the fatalities and \$19 billion of the damages occurred only in NYC (City of New York, 2013). Records showed that parts of the Battery Park in Lower Manhattan were inundated with up to 3 meters of seawater (Blake et al., 2013).

2.2. Human Stability Model

A human stability model is developed here to account for drowning and toppling and slipping instability of individuals exposed to coastal flooding accompanied by extreme winds. The model considers drowning to occur when the water surface level is above an individual's chin. The model considers toppling and slipping stability failures induced by both floodwater and wind. Winds are assumed to blow toward the individual from an angle of β_{wr} relative to the floodwater flow direction (Figure 2). The ground slope, relative to the horizontal plane, is θ in the floodwater direction. The ground slope is γ in the upwind direction. Like Lind et al. (2004) and Milanese et al. (2014), we parameterize the human body as a set of cylinders that stand in the vertical position, facing the approaching floodwater flow (Figure 2). These assumptions are typically adopted in human stability models. Considering an individual facing the approaching floodwater is more cautionary because the pivot point (of toppling) is represented by the heel and, thus, the moment arm of the stabilizing force is minimal (Milanese et al. 2014). For a human body with a height of Y , legs are schematized by two paired cylinders of diameter d and height of $Y/2$. The torso is modeled as a single cylinder with a diameter of $D = 2d$ and height of $Y/2$.

In Figure 2, the longitudinal axis η represents the axis from the upstream (the direction the floodwater comes from) to downstream, which makes an angle of θ with the horizontal plane. The normal axis ξ is the

axis normal to the longitudinal axis η . The lateral axis ζ is parallel to the horizontal plane and perpendicular to the longitudinal axis.

2.2.1. Slipping Instability

The instability due to slipping lies on the equilibrium of driving and resisting forces acting on the human body. The model considers an individual to be physically stable if the resultant driving force acting on the body is equal or smaller than the resultant resisting force. According to Milanesi et al. (2014), the driving force represents the resultant drag force induced by floodwater flows. Here we further include the drag force induced by winds. The resisting forces are calculated as the product of a friction coefficient (between the sole and the ground) and the individual's effective weight. The effective weight is calculated as the summation of forces in the normal direction ξ . The individual becomes physically unstable if the following inequality between the magnitude of driving (left-hand side of the inequality) and resisting (right-hand side) forces is met

$$\sqrt{(W_\eta + F_{h,\eta} + F_{a,x,\eta})^2 + (F_{a,y})^2} > \mu(W_\xi - B - F_{h,\xi} - F_{a,x,\xi}) \quad (1)$$

where W_η and W_ξ are, respectively, the longitudinal and normal components of the bodyweight W and are calculated as $W\sin(\theta)$ and $W\cos(\theta)$; $F_{h,\eta}$ and $F_{h,\xi}$ are the longitudinal and normal components of the hydrodynamic force F_h and are calculated as

$$F_{h,\eta} = F_h \cos(\theta) = \frac{1}{2} \rho_f C_f [U_f \cos(\theta)]^2 A_f \cos(\theta) \quad (2)$$

$$F_{h,\xi} = F_h \sin(\theta) = \frac{1}{2} \rho_f C_f [U_f \cos(\theta)]^2 A_f \sin(\theta) \quad (3)$$

where U_f is the floodwater speed along the longitudinal axis, ρ_f is the floodwater density ($1,017 \text{ kg.m}^{-3}$ for seawater), C_f is the floodwater drag coefficient for a circular cylinder, and A_f is the frontal area of the body which is perpendicular to the floodwater direction (see Supporting Information). Note that dynamic forces act perpendicular to the frontal area and, thus, F_h is calculated using the horizontal component of the floodwater speed, that is, $U_f \cos(\theta)$.

In Equation 1, $F_{a,x,\eta}$ and $F_{a,x,\xi}$ are, respectively, the longitudinal and normal components of the aerodynamic force (after the force is decomposed to x and y directions), and calculated as $F_{a,x} \cos(\theta)$ and $F_{a,x} \sin(\theta)$. The x and y components of the decomposed aerodynamic force are

$$F_{a,x} = F_a \cos(\beta_{wr}) = \frac{1}{2} \rho_a C_w [U_w \cos(\gamma)]^2 A_w \cos(\beta_{wr}) \quad (4)$$

$$F_{a,y} = F_a \sin(\beta_{wr}) = \frac{1}{2} \rho_a C_w [U_w \cos(\gamma)]^2 A_w \sin(\beta_{wr}) \quad (5)$$

where U_w is the wind speed (along the upwind-downwind direction), ρ_a is the air density (1.20 kg.m^{-3}), C_w is the wind drag coefficient for a circular cylinder, and A_w is the frontal area of the body which is perpendicular to the wind direction (see Figure S3 and Supporting Information).

Like Milanesi et al. (2014), the buoyancy force is calculated as $B = \rho_f g V_s \cos(\theta)$, where g is the gravitational acceleration ($9.81 \text{ m}^2.\text{s}^{-1}$), and V_s is the submerged volume of the body (see Supporting Information). The buoyancy force acts in the direction normal to the bed (Armanini & Gregoretti, 2005). The friction coefficient μ is set to 0.46 as used by Milanesi et al. (2014).

2.2.2. Toppling Instability

Toppling instability is evaluated based on the moment of forces about the pivot point of the body. This approach was used by Milanesi et al. (2014). Here we further include the moment of forces due to winds. The heel, point P labeled in Figure 2, is considered as the pivot point in the longitudinal axis η . Toppling in the longitudinal axis occurs if the following inequality is met.

$$F_{h,\eta} \zeta_{F_h} + F_{h,\xi} \eta_{F_h} + F_{a,x,\eta} \zeta_{F_{a,x}} + F_{a,x,\xi} \eta_{F_{a,x}} + W_\eta \zeta_W + B \eta_B > W_\xi \eta_W \quad (6)$$

Table 1
Human Body Characteristics Used in the Present Study

Body characteristic, notation (units)	Children	Adults
Mass, m (kg)	22.4	81.6
Height, Y (m)	1.21	1.69
Torso diameter, D (m)	0.17	0.28
Leg diameter, d (m)	0.085	0.14
Wind Drag Coefficient, C_w	1.4	1.4
Water Drag Coefficient, C_f	0.9	0.9
Friction Coefficient, μ	0.46	0.46

where ξ and η with a subscript represent the moment arms about the pivot point P, that is, the perpendicular distance between the acting location of a force and the pivot point. The moment arms are calculated based on the locations where forces act. The weight acts on the center of mass of the body, and the buoyancy force acts on the center of gravity of the body's submerged volume. The acting locations of the hydrodynamic and aerodynamic forces are calculated using the assumption that dynamic pressures are uniformly distributed on the frontal body area. Table S1 in Supporting Information provides formulations of the moment arms.

Toppling in the direction of the lateral axis ζ occurs due to the buoyancy and aerodynamic forces. The hydrodynamic force does not cause toppling in the lateral direction because of the assumption that the individual faces the floodwater flow. Thus, hydrodynamic forces are along the longitudinal axis. The outer edge of the foot, that is point Q in Figure 2,

is considered as the pivot point in the lateral direction. Toppling instability in the lateral direction occurs when the following inequality is satisfied

$$F_{a,y}z_{F_{a,y}} + B_z y_{B_z} > W y_W \quad (7)$$

where B_z is the vertical component of the buoyancy force ($B \cos(\theta)$), and z and y with a subscript represent the moment arms (see Supporting Information).

2.2.3. Body Characteristics

We utilize the developed model to study the vulnerability of both adults and children to coastal flooding. To quantify forces acting on the body, the model requires a series of parameters that represent the human body characteristics, including mass, height, and dimensions. Based on the best available data in the literature, we set the body characteristics to values summarized in Table 1. These values represent the worldwide-average characteristics of 7-year-old children (C. D. Fryar et al., 2012; Milanese et al., 2014) and the average characteristics of the U.S. adults of age range from 20 to 60 (C. Fryar et al., 2018).

2.2.4. Categories of Vulnerability

We categorize flooded areas, in terms of the type of human vulnerability to coastal flooding, into drowning, failure, hazardous, and safe zones (Table 2). A drowning zone represents flooded areas where the water level is above the chin, that is, $h > \frac{13Y}{16}$ where h is the flood depth and Y is the height of the individual.

Failure zones represent flooded areas where an individual is subject to physical instability due to slipping or toppling, that is when either Equations 1, 6, or 7 is satisfied. A flooded area is categorized as a hazardous zone, if none of Equations 1, 6, or 7 is satisfied. It indicates that although neither drowning nor physical instability occurs, the floodwater in the region can be hazardous (and may cause instability to individuals with a weak physical strength).

Areas which remain dry (i.e., no floodwater) during the flood event are categorized as safe. Note that dried areas are still subject to strong winds during storms. However, this study focuses on combined hazards from floodwater and wind and, thus, excludes areas that are solely subject to wind hazards.

We further quantify the degree of hazard in hazardous zones using a Hazard Rating index (HR). Kvocka et al. (2016) developed the following hazard rating index, HR, which ranges between 0 and 1, to define four hazard degrees (low, moderate, significant, and extreme), with three threshold values, 0.3, 0.6, and 1.

Table 2
Categories of Vulnerability of Human to Coastal Flooding

Type of vulnerability	Conditions
Drowning	$h > (13Y / 16)$
Failure	Toppling Instability Equations 6 or 7
	Slipping Instability Equation 1
Hazardous	High Hazard $HR \geq 0.6$
	Medium Hazard $0.3 \leq HR < 0.6$
	Low Hazard $0 < HR < 0.3$
Safe	$h = 0$

$$HR = \min \left[1, \frac{U_f}{\min[U_{f,cs}(h), U_{f,ct}(h)]} \right] \quad (8)$$

where $U_{f,cs}(h)$ and $U_{f,ct}(h)$ are, respectively, the critical floodwater speed in which the slipping and toppling instabilities in the longitudinal direction occur for the given water depth h . U_f is the instantaneous floodwater speed. The minimum value between $U_{f,cs}$ and $U_{f,ct}$ represents the critical floodwater speed $U_{f,c}$. This hazard rating index does not consider the influence of the wind. Winds, depending on the wind direction, change the critical floodwater speeds $U_{f,cs}$ and $U_{f,ct}$ and may also cause toppling in the lateral direction. Here we modify the formula of Kvočka et al. (2016) to develop a hazard rating index that accounts for the effects of winds on $U_{f,cs}$ and $U_{f,ct}$ and also toppling in the lateral direction. The modified HR is formulated as

$$HR = \min \left[1, \max \left(\frac{U_f}{\min[U_{f,cs}(h, \beta_{wr}, U_w), U_{f,ct}(h, \beta_{wr}, U_w)]}, \frac{U_w}{U_{w,ct}(h, \beta_{wr})} \right) \right] \quad (9)$$

where $U_{f,cs}(h, \beta_{wr}, U_w)$ and $U_{f,ct}(h, \beta_{wr}, U_w)$ are, respectively, the critical floodwater speed in which the slipping and toppling instabilities in the longitudinal direction occur for given water depth h , wind relative angle β_{wr} , and wind speed U_w . The minimum value between $U_{f,cs}$ and $U_{f,ct}$ represents the critical floodwater speed $U_{f,c}$. In our HR formula, $U_{w,ct}(h, \beta_{wr})$ is the critical wind speed in which the toppling instability in the lateral direction occurs for the given water depth and relative angle. Using the HR Formula 9, we subcategorize the hazardous zones into three zones: low, medium, and high, with two threshold values, 0.3 and 0.6 (Table 2).

2.3. Hydrodynamic and Wind Data

The human stability model calculates the hydrodynamic and aerodynamic forces based on the floodwater and wind characteristics including the water depth and velocity components, and wind velocity components. We obtain the floodwater and wind characteristics during Hurricane Sandy from a high-resolution flood model of Manhattan, NY, developed by Marsooli and Wang (2020). The flood model is based on the ADvanced CIRCulation (ADCIRC) model (Luettich et al., 1992; Westerink et al., 1994) coupled with the Simulating WAVes Nearshore (SWAN) model (Booij et al., 1999; Dietrich et al., 2011). ADCIRC is a finite element coastal ocean model that solves the depth-integrated continuity and mean flow momentum equations to calculate the water depth and velocities. SWAN is a spectral wave model that solves the depth-integrated wave action balance equation to calculate the wave action density and, in turn, the phase-averaged wave characteristics. Coupling ADCIRC with SWAN allows the model to include the effects of wind waves and swells on the water depth and velocities.

The computational mesh in Manhattan has a spatial resolution of 3–8 m and resolves urban features such as buildings and roads. The computational domain in Manhattan extends to an elevation of 10 m above the mean sea level. Marsooli and Wang (2020) implemented the model to simulate coastal flooding in Manhattan during Hurricane Sandy. The comparisons between the measured and modeled flood depths (i.e., high-water marks) in Manhattan showed a root-mean-square-error of 16 cm and a bias of 5 cm. In the model, the wind forcing was based on the Reanalysis data from Oceanweather Inc. (www.oceanweather.com). The ADCIRC model implemented a directional land-masking approach (Bunya et al., 2010; Westerink et al., 2008) to account for the friction of land features, for example, buildings and trees, on the wind speed. Wind data in the present study are the outputs from ADCIRC. More details on the flood modeling can be found in Marsooli and Wang (2020).

3. Model Calibration

We calibrate the floodwater and wind drag coefficients, C_f and C_w respectively, in our human vulnerability model using existing laboratory measurements of critical water and wind speeds. Due to a lack of measurements under conditions that floodwater and wind coexist, we separately calibrate the floodwater and

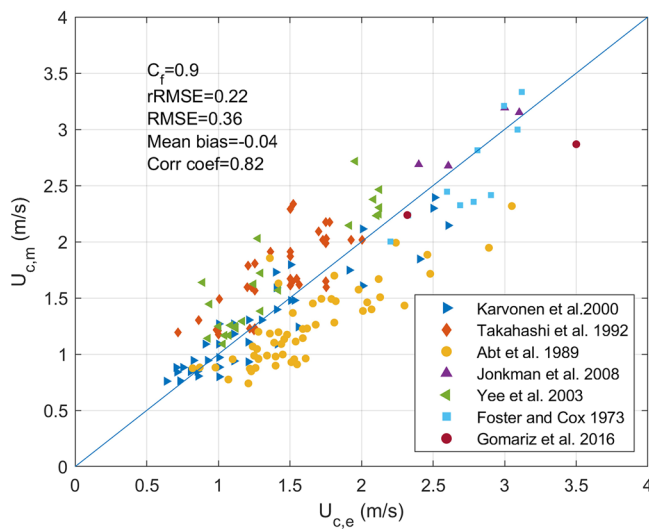


Figure 3. Comparison between modeled ($U_{c,m}$) and measured ($U_{c,e}$) critical floodwater speed. Model results are based on a floodwater drag coefficient of 0.9.

that the model with $C_f = 0.9$ produces critical floodwater speeds that have the best agreement with measurements (Figure 3). The model has a RMSE of 0.35 m/s, rRMSE of 0.22, mean bias of -0.04 m/s, and correlation coefficient of 0.82.

Compared to laboratory measurements of human instability due to floodwaters, a literature review shows that measurements of human instability due to winds are scarce. Therefore, we are only able to calibrate the model's wind drag coefficient C_w using a limited number of measurements. Jordan et al. (2007) performed a series of laboratory experiments to measure the displacement of individuals exposed to winds of different speeds. The average mass and height of subjects were, respectively, 72.2 kg and 1.81 m. We determine the measured critical wind speed as the speed that initiates the displacement of a subject. Model calibration indicates that the model with $C_w = 1.4$ produces the most accurate results, with a RMSE of 3.18 m/s, rRMSE of 0.24, and mean bias of 0.20 m/s.

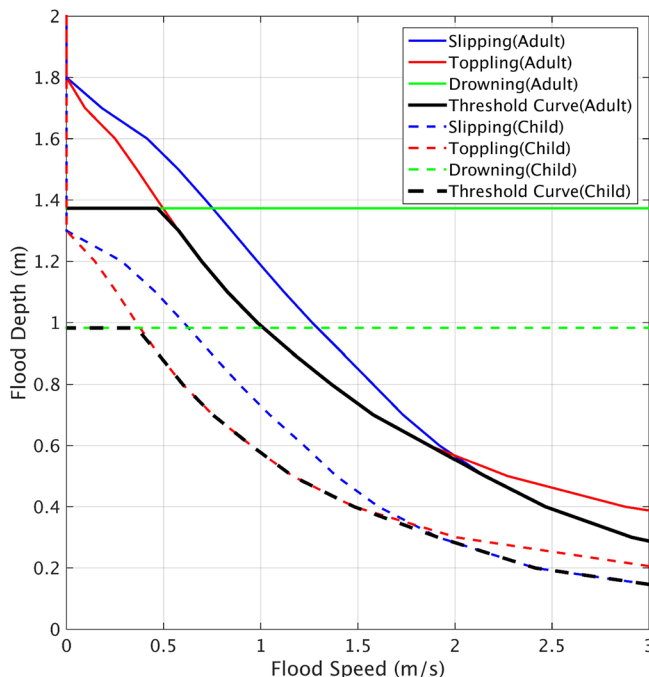


Figure 4. Human body instability threshold curve for adults and children without wind effect.

wind drag coefficients. The goodness of fit between the model results and laboratory measurements is quantified using root-mean-square-error (RMSE), relative RMSE, mean bias, and correlation coefficient. The relative RMSE is calculated as the following

$$\text{rRMSE} = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (U_{c,m} - U_{c,e})^2}}{\overline{U_{c,e}}} \quad (10)$$

where $U_{c,m}$ and $U_{c,e}$ are, respectively, the modeled and measured (experimental) critical speed, n is the number of data points, and the overbar represents the mean.

The floodwater drag coefficient C_f is calibrated using measurements from laboratory experiments conducted by Abt et al. (1990), Karvonen et al. (2000), Takahashi et al. (1992), Yee (2003), Foster and Cox (1973), Jonkman and Penning-Rowsell (2008), and Martínez-Gomariz et al. (2016). The experiments measured the critical water velocity, under given water depths, that caused human subjects to become physically unstable. The subjects in the experiments consisted of adults and children with a mass between 25 and 100 kg and a height between 1.09 and 1.95 m (Table S3). In total, we used 176 measurements, which include 33 measurements for children and 143 for adults. Model calibration shows

4. Results

This section first presents the human instability threshold curves and next the vulnerability of individuals to coastal flooding in Manhattan during Hurricane Sandy.

4.1. Human Instability Threshold Curves

For a specific floodwater depth (or speed), in the absence of winds, physical instability occurs when the floodwater speed (or depth) exceeds a critical value, that is, a threshold. Here, we determine the critical flood depth (or speed) for a given floodwater speed (or depth) using the instability conditions formulated in the Methodology section. Figure 4 shows an example of threshold curves for drowning and slipping/toppling instabilities of adults and children in the absence of wind (and for a ground slope of zero). The figure also shows the failure curve, which integrates the threshold curves for drowning and slipping/toppling instabilities.

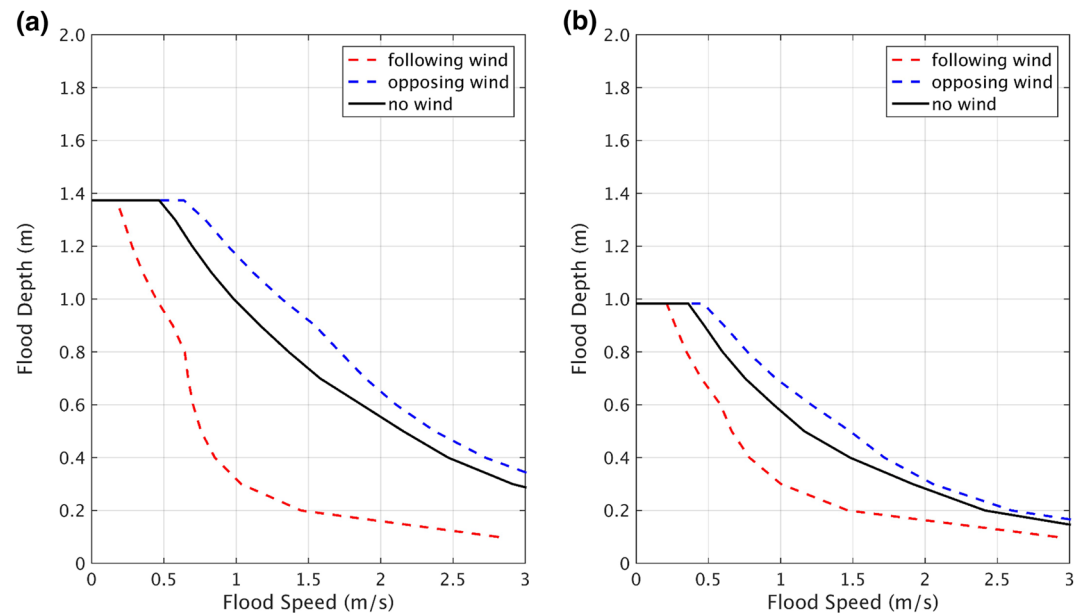


Figure 5. Wind effects on human instability threshold curve for (a) adults and (b) children. The wind speed is set to 15 m/s for an adult and 10 m/s for a child.

According to the threshold curve for an adult individual, for a floodwater speed smaller than 0.47 m.s^{-1} , physical instability occurs when the flood depth is larger than the critical depth of drowning. Therefore, drowning occurs before slipping or toppling. For floodwaters with velocities higher than 0.47 m.s^{-1} , either slipping or toppling instability occurs prior to drowning. For instance, for a floodwater speed of 1 m.s^{-1} , the failure occurs due to toppling instability (at a flood depth of 0.99 m), and for a floodwater speed of 2.5 m.s^{-1} , the failure occurs due to slipping instability (at a flood depth of 0.39 m). The threshold curves indicate that as the floodwater speed increases, the critical depth decreases, and the slipping instability becomes the cause of failure. The threshold curves for a child show smaller critical depths and speeds. This is because the light weight and short height of children result in smaller resisting forces compared to flood-induced driving forces.

Depending on the wind direction, winds would escalate or de-escalate the physical instability of an individual exposed to floodwaters. To demonstrate the wind effect on the threshold curve, two extreme conditions occurring when winds blow either in the same or opposite direction as the floodwater flow, that is, following or opposing winds, are taken as examples (Figure 5). Figure 5a shows the effect of following and opposing winds (with a wind speed of 15 m.s^{-1}) on adults' stability threshold curves. For a given floodwater depth (or speed), the following winds result in a substantial reduction in the floodwater critical speed (or depth). For instance, for an adult exposed to flooding with a flood depth of 0.8 m , the critical floodwater speed that results in physical instability is 1.36 m.s^{-1} in the absence of winds but it is only 0.64 m.s^{-1} in the presence of the following winds. In contrast, opposing winds generate forces applied to the body in the opposite direction of the floodwater flow, increasing the critical floodwater depth and speed and, in turn, improving the physical stability. For example, for the given flood depth of 0.8 m , the opposing wind increases the critical water speed from 1.36 to 1.72 m.s^{-1} .

4.2. Vulnerability to Hurricane Sandy Flood Hazards

Results from the hydrodynamic model indicate that Hurricane Sandy caused coastal flooding over a large area near the shoreside of Lower Manhattan (Figure 6). The maximum flood depth reached up to 3 m locally in some neighborhoods (4.1% of the total flooded area), for example, the East Village area and the Two Bridges area on the east side (see Figure 1 for the location of these areas). The maximum floodwater speed was below 0.5 m/s in 81% of the total flooded area in the study domain (Figure 6). However, the maximum

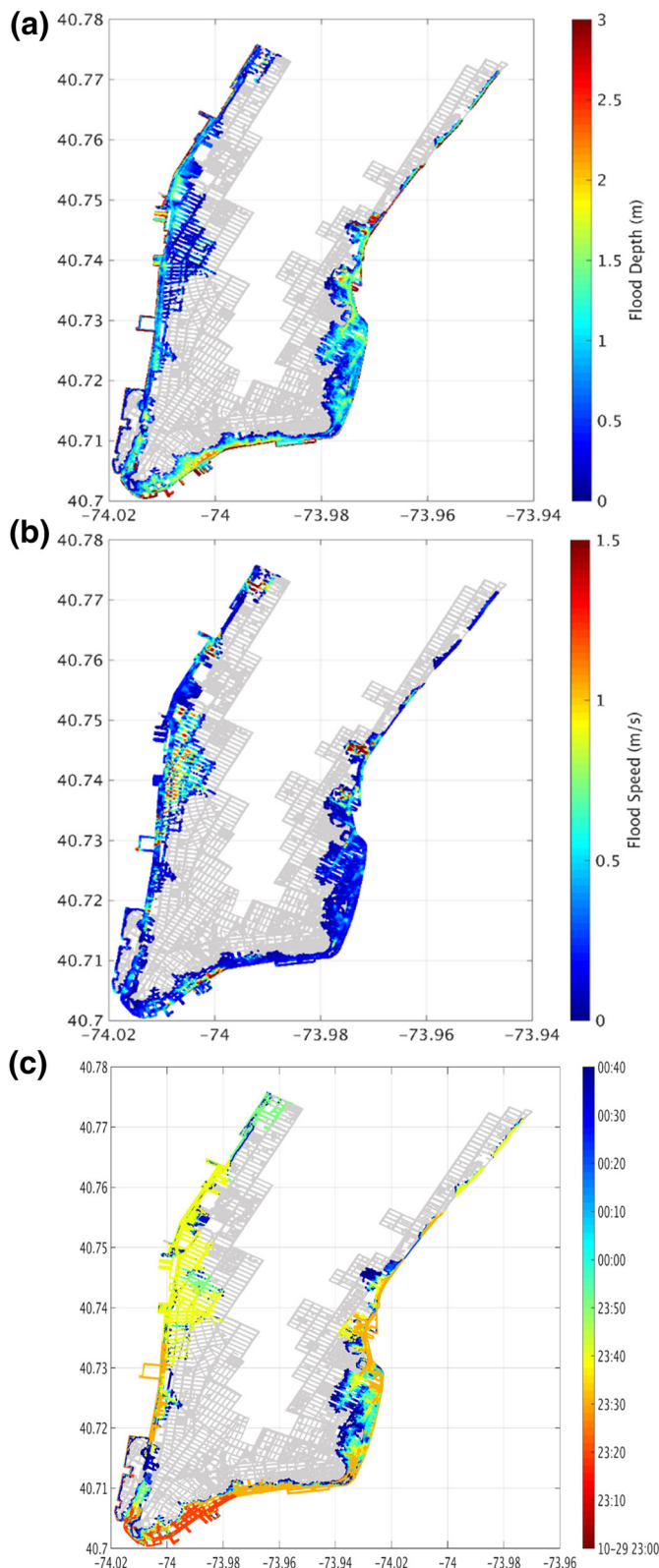


Figure 6. Spatial-distribution of the simulated (a) maximum floodwater depth, (b) maximum floodwater speed and (c) flood arrival time in Manhattan during Hurricane Sandy. Times are in Greenwich Mean Time Zone.

water speed reached up to 1.5 m/s in some localized areas (1.4% of the total flooded area), for example, the north of the East Village, the north of the West Village, and the Two Bridges area. Individuals exposed to deep, fast-moving floodwaters in these areas would be highly vulnerable to failure and drowning.

Simultaneous occurrence of deep, fast-moving flood conditions in multiple neighborhoods could impact rescue of stranded individuals, especially when a large area is flooded within a short period of time. Hydrodynamic model results show that flooding arrived first in the southern part of Manhattan and within 20 min large areas on the east and west sides of Manhattan were flooded (Figure 6). The east side of Manhattan (along the East River) was flooded slightly earlier than the west side (along the Hudson River).

The human stability model applied to Hurricane Sandy considers winds as a stochastic variable in space and time. Spatially distributed time series of wind speed and direction are based on outputs of the ADCIRC + SWAN model at each computational node. Results reveal that most flooded areas in Manhattan were, at their worst condition during Sandy, mainly a hazardous zone for adults and either a failure or drowning zone for children (Figure 7). Hazardous zones posed low-to-high hazards to adults but mainly high hazards to children. At the worst floodwater and wind condition during Sandy, about 5.35% and 47.4% of the total flooded area became a failure zone for adults and children, respectively, and 19.9% and 42.4% became a drowning zone. Most drowning zones were in the southern and eastern neighborhoods of Manhattan, consistent with the deep flood depth in these neighborhoods (Figure 6). Some flooded areas that became a drowning zone reside critical facilities such as hospitals and fire stations. For example, Figures 7b and 7d show that a wide area near the Bellevue Hospital, located in the east side of Manhattan, became a drowning zone.

Figure 8 shows that large portions of the flooded areas remained a failure or drowning zone for a duration of longer than an hour. Among the flooded areas that became a drowning zone during Hurricane Sandy, 14.1% and 31.5% for adults and children, respectively, remained a drowning zone for a duration of longer than an hour. The long duration of the drowning condition is due to the localized topography and terrain slope that result in water accumulation in low-lying neighborhoods. Compared to drowning zones, we found that the estimated duration that flooded areas remained as a failure zone is relatively short (Figure 9). For example, for children, while the duration of drowning in 50% of drowning zones was below 100 min, the duration of failure in 50% of failure zones was below 40 min. The short duration of failure is because of the rapid momentum loss of floodwater when flood waves propagate over land. Failure condition, due to either toppling or slipping instability, occurs when the floodwater speed surpasses a critical value (for a given flood depth). In coastal urban environments where the land is characterized by a gentle slope, the floodwater rapidly loses its momentum (to friction) as it propagates further inland.

The model results show that winds substantially affected the physical vulnerability of individuals to coastal flooding during Hurricane Sandy, especially for children. Figure 10 shows the percentage of flooded areas that became a hazardous or failure zone, during the peak floodwater and

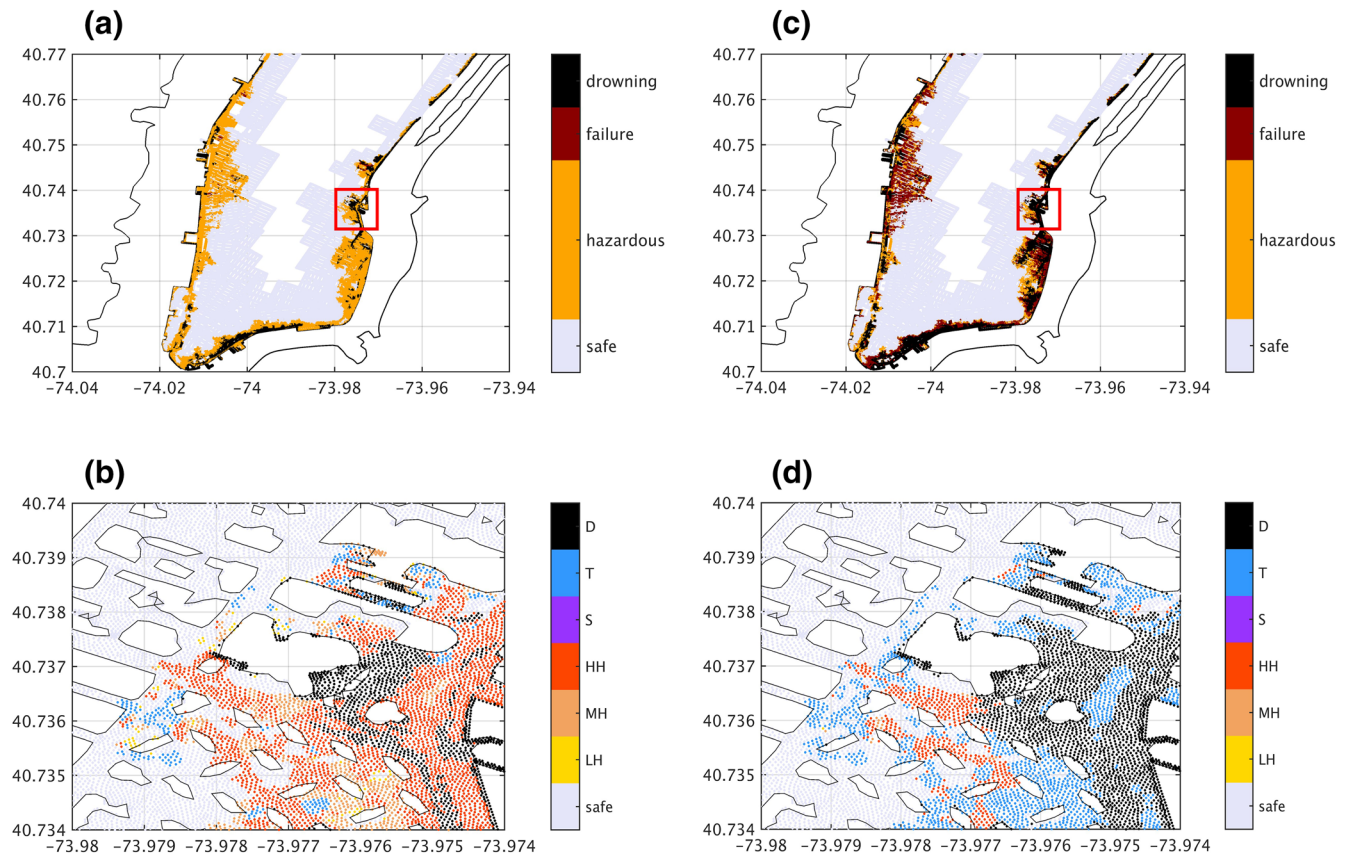


Figure 7. Spatial distribution of the highest level of vulnerability for (a–b) adults and (c–d) children during Hurricane Sandy (upper panels for the entire study area; lower panels for a neighborhood where the Bellevue Hospital is located). Lower panels further categorize the hazard type to low, medium, and high hazard levels and the failure type to toppling and slipping (D: drowning; T: toppling; S: slipping; HH: high hazard; MH: medium hazard; LH: low hazard).

wind hazards, in the presence and absence of winds. Overall, wind effects make some flooded areas that are hazardous zones to become a failure zone, which can be seen by a reduction in the area of hazardous zones but an increase in the area of failure zones. For instance, for children, while the area of hazardous zone decreases from 2.302 to 0.417 km² (81.9% reduction), the area of failure zone increases from 0.045 to 1.930 km² (4188.4% increase).

The simulated time series of human vulnerability across the study area indicates that wind effects lead to an increase in the time duration that a flooded area remained as a hazardous or failure zone during Hurricane Sandy. For example, for children and in the absence of wind effects, at a location near the Bellevue Hospital Center (Figure 11), the site first becomes a drowning zone for a short period of time and then becomes a hazardous zone with a low to medium hazard level. In the presence of wind effects, on the other hand, the flooded area becomes a drowning and failure zone for a period of time and then becomes a hazardous zone with a hazard level that varies between low and high. Neglecting wind effects at this location results in underestimating the human vulnerability for a total of 6.2 h for adults and 13.3 h for children.

5. Discussion

This study developed a physics-based model to predict the individuals' physical vulnerability to combined hazards due to floodwater depth and speed and winds. The model can be utilized by emergency management and response organizations to identify the location and timing of flooded areas with different vulnerability levels, from (low/medium/high) hazardous to failure and drowning. If coupled with flooding forecasting systems that predict hazards (due to water and wind), the vulnerability model can be used for

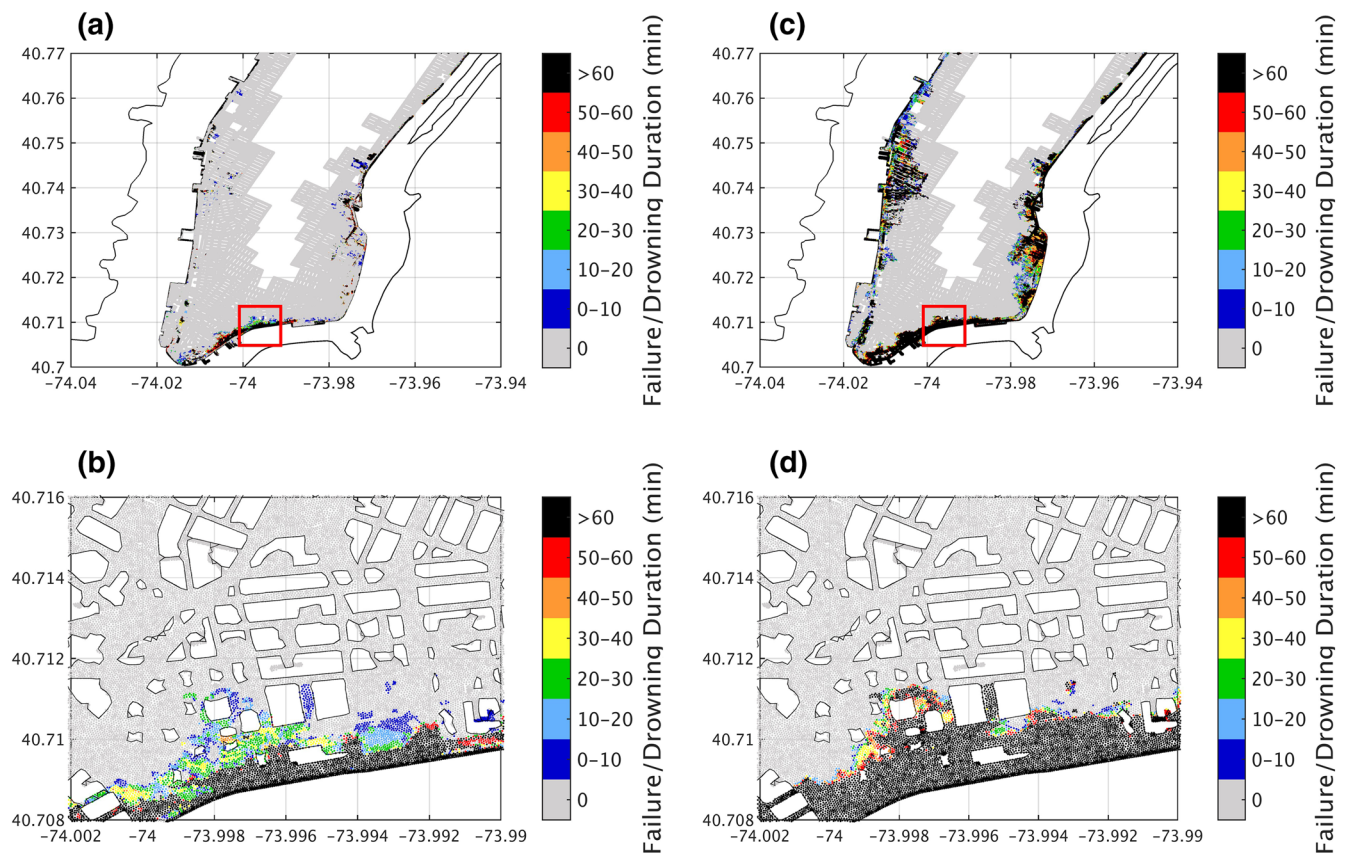


Figure 8. Spatial distribution of the time duration that flooded areas remained either a failure or drowning zone. Left panels are for adults, and the right panels are for children.

preparedness and response planning for a forecasted coastal flood event. For example, vulnerability maps can be generated to inform transport facilities management and safe and effective evacuation and rescue operations. The model could also be adopted in life loss models, for example HEC-LifeSim, to determine the physical instability of individuals based on the combined hazards due to floodwater and winds.

Hazards and vulnerability are two integral components in flood risk assessment. In the built environment, hazards associated with coastal storms such as hurricanes consist of multiple components, including flood depth, floodwater speed, and wind speed. Consequently, the vulnerability of communities and assets to coastal storms should account for the coexistence of multi-hazards. Flood hazard assessment in coastal areas by characterizing flood hazards based on a single hazard component, for example the water level with different return periods, could underestimate hazards and vulnerability. To generate flood maps, the water level of different return periods is usually mapped, using a static approach called “bath tubbing” into coastal areas to determine the extent and depth of flooding. Vulnerability assessment based on a single hazard, that is, the water level, would mask the actual vulnerability of humans and assets to coastal flooding. Results from this study suggest that coastal flood hazard and vulnerability assessments should account for combined hazards due to flood depth, floodwater speed, and wind.

The present study was based on the analysis of driving and resisting forces applied to the human body to determine the vulnerability of individuals to flooding while neglecting the effects of other environmental and human factors. We assumed that the individual exposed to flooding stands upright and faces the approaching floodwater. Future studies should investigate how facing different angles and adjusting the posture would influence an individual’s physical stability. For example, an individual may better resist toppling if squat down because of the lower center of mass. Maneuverability could also affect the individual’s stability due to the instantaneous changes in the center of mass and the frontal area exposed to drag forces.

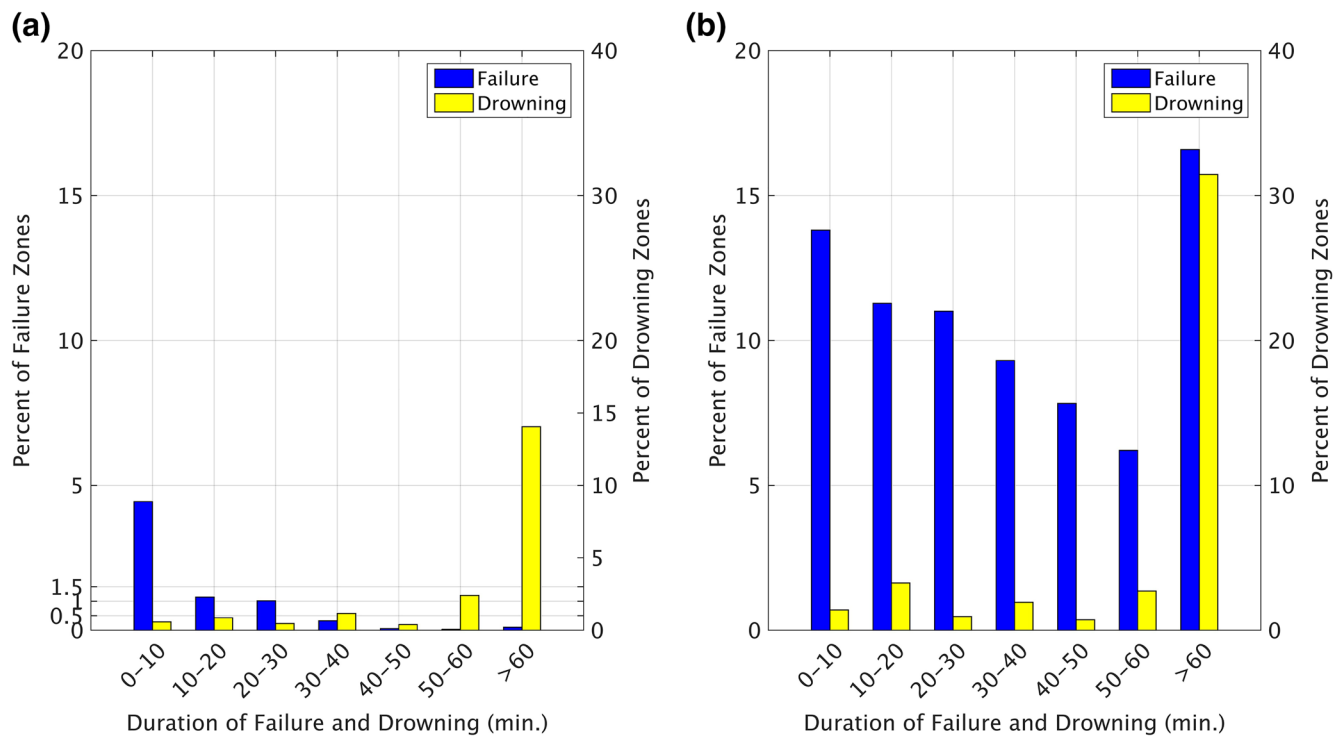


Figure 9. The percentage of the total flooded area that became a failure or drowning zone, for a specific time duration, for (a) adults and (b) children.

New methods should be developed to properly incorporate the natural human maneuverability into the stability models. Among other human factors that could impact the human vulnerability are the psychological attributes and physique of the individual exposed to flooding. Environmental factors such as wind gust, flow

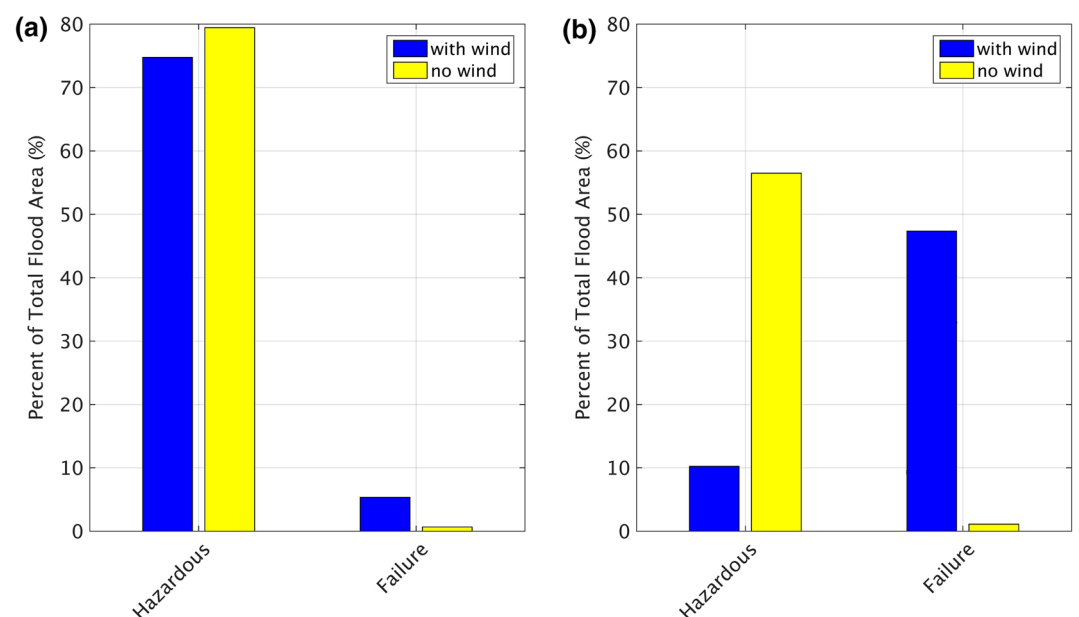


Figure 10. Percentage of total flooded area that became a hazardous or failure zone, at the worst floodwater and wind condition, when wind effects are included (blue) and neglected (yellow) for (a) adults and (b) children.

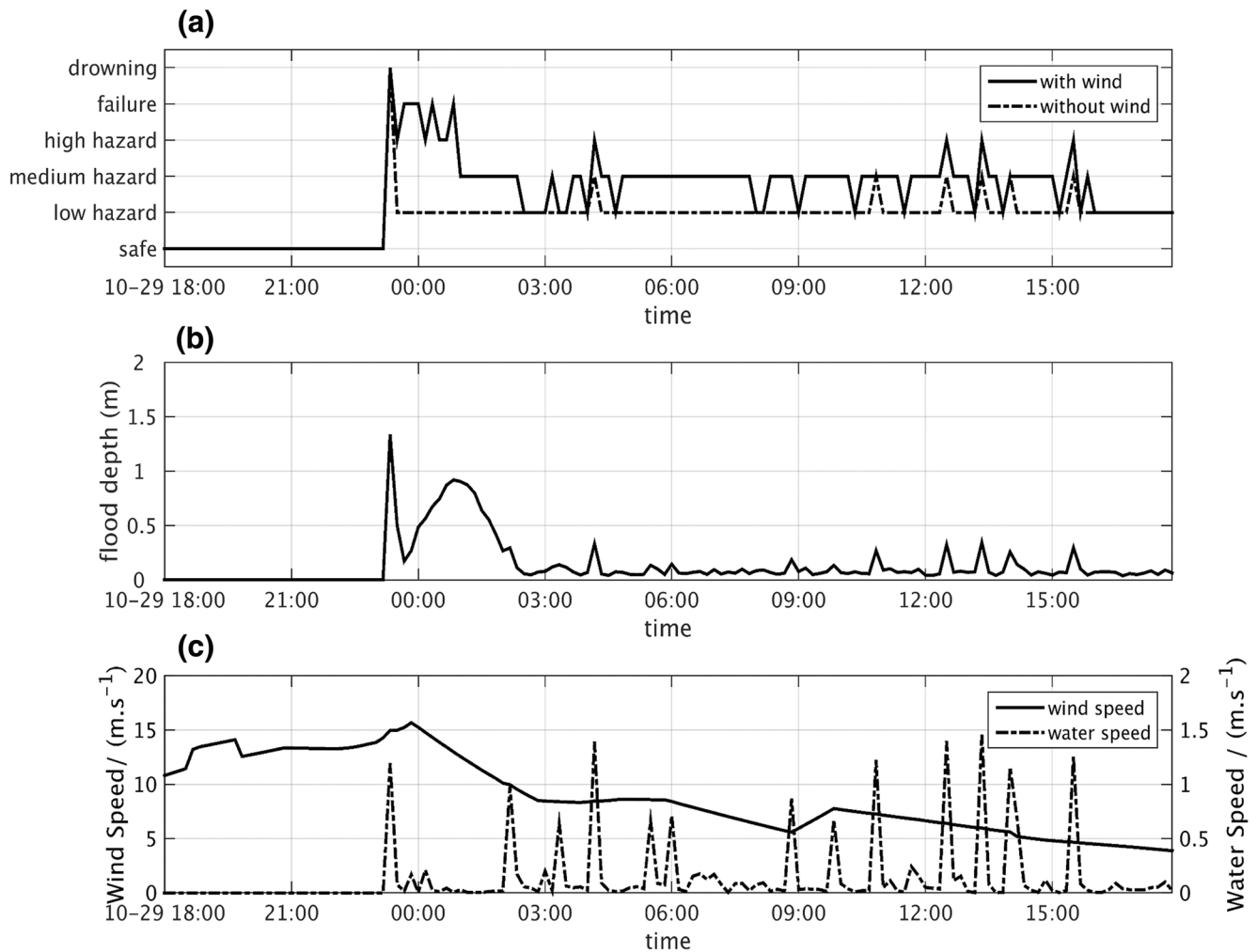


Figure 11. Time series of (a) hazard type, (b) flood depth, and (c) floodwater speed and wind speed at a location near Bellevue Hospital Center. The hazard types are estimated for children.

turbulence, floating debris, among others, as well as early warning systems, can also affect the vulnerability of humans to coastal flooding.

The model performance was evaluated by comparing the critical floodwater speed (that initiates human instability) calculated by the model with those measured in a series of existing laboratory experiments (Figure 3). Model validation against field data is challenging because of a lack of hydrodynamic measurements and human instability observations during historical flood events, including Hurricane Sandy. Future field studies are needed to collect floodwater and wind velocity measurements in the built environment during coastal flood events. Human instability datasets may also be generated based on surveillance camera footage of pedestrians' instability incidents in flooded areas.

The developed model requires three predefined parameters, that is bottom friction, water drag, and wind drag coefficients. The friction coefficient was set to the value reported in the previous studies. The bottom friction is a function of the pavement surface and footwear type. While pavement surface in the built environment is usually constructed of a rigid hard surface (e.g., concrete and asphalt), the type of footwear can widely vary, which results in uncertainties in model inputs. Water and wind drag coefficients were calibrated by comparing model outputs against a series of existing laboratory measurements. Model calibration was separately carried out for the water and wind drag coefficients due to the lack of measurements when

individuals are concurrently exposed to both floodwater and wind. Future laboratory/field studies are needed to investigate the physical instability of individuals when floodwater and wind coexist.

Human stability models contain a series of uncertain input data including the human body characteristics (i.e., weight, height, and size), outfit type (i.e., loose or tight clothes, type of fabric), and footwear type (boots, shoe, and sandal). In the present model, we followed a deterministic approach and adopted a single value for each input parameter. To adopt such models for decision-making, however, it is necessary to properly account for the uncertainties in model inputs. Therefore, the models should be implemented within a probabilistic/ensemble framework in which various combinations of input scenarios are considered.

6. Conclusion

We developed a human vulnerability model to evaluate the physical vulnerability of individuals to coastal flooding induced by windstorms. The model predicts the physical instability of an individual, who is concurrently exposed to floodwaters and winds, based on the driving and resisting forces and moments applied to the body. The model categorizes the flooded areas to hazardous, failure, and drowning zones, and accounts for the failure caused by physical instability due to both slipping and toppling. Hazardous zones are considered as flooded areas where while neither failure nor drowning occurs, the coexistence of floodwater and wind results in low, medium, or high hazards. The model parameters, that is drag coefficients, were calibrated using exiting laboratory data.

We utilized the model to generate human instability threshold curves that determine the floodwater critical depth (or speed) that results in slipping or topping failure. The modeled threshold curves showed that, depending on the wind direction, winds could escalate or de-escalate individuals' physical instability exposed to floodwaters. With a focus on two extreme wind direction scenarios (following and opposing winds, relative to the floodwater flow direction), we found that the threshold curve is significantly different in the absence and presence of winds. The following wind decreases the critical water speed for a given water depth while the opposing wind helps the human body to resist the floodwater loading and, thus, contributes to stability. For example, for an adult exposed to a flood depth of 1 m, while the critical speed is 0.98 m/s in the absence of winds, it is 1.32 and 0.45 m/s in the presence of opposing and following winds (with a speed of 15 m/s), respectively.

The model application to Hurricane Sandy showed that most flooded areas in Manhattan became a hazardous zone (74.75% of the total flooded area) for adults and either a failure (47.4% of the flooded area) or drowning zone (42.4% of the flooded area) for children. Large portions of the flooded areas remained as a failure or drowning zone for longer than an hour. The duration that flooded areas remained as a failure zone was relatively shorter than the duration that they remained as a drowning zone. This can be because of a rapid loss of floodwater momentum (and thus a rapid reduction in the water velocity) when flood waves propagate over land. The model results showed that winds significantly increased individuals' vulnerability to coastal flooding during Sandy, especially for children. Neglecting wind effects resulted in a reduction of 97.7% in the area of failure zones for children. Besides, the results pointed out that winds increased the duration that flooded areas remained as a failure zone or a hazardous zone with a higher hazard level. For example, near a critical facility in the study area, that is, the Bellevue Hospital Center, the effect of winds on the time duration that flooded areas remained hazardous and failure zones was to increase it by 6.2 h for adults and 13.3 h for children.

The developed vulnerability model can be coupled with hazard models for use in flood risk assessment studies. The model can also be coupled with flood forecasting systems to inform advanced flood preparedness and emergency response plans. The developed model can be expanded to account for other environmental and human factors that influence the individuals' vulnerability to coastal flooding induced by windstorms, including wind gust, flow turbulence, floating debris, and psychological attributes and physique of individuals.

Data Availability Statement

Data used for calibrating the human stability model can be accessed from the cited literature papers. The hydrodynamic data for Hurricane Sandy were obtained from Marsooli and Wang 2020.

Acknowledgments

Supporting Information contains all the calculation formulas and schematic diagrams of the Human Stability Model. The authors thank three anonymous reviewers for their constructive comments.

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